CMT 442 – Neural Networks

Course Description

This course introduces neural networks as a core component of artificial intelligence, covering theoretical foundations, learning algorithms, deep learning architectures, and responsible AI governance aligned to international standards.

Course Objectives / Learning Outcomes (CBET)

- Design and implement neural network models for real-world problems
- Evaluate performance, bias, explainability, and risk in AI systems
- Apply responsible AI and governance principles in neural network development

Course Structure and Standards Mapping (6 Modules)

Module 1: Foundations of Neural Networks and AI

Indicative Content:

- History of AI
- Biological inspiration
- Neural network basics

Mapped Standards / Frameworks:

- ISO/IEC 42001:2023 – AI context and principles
- NIST AI RMF – Govern

Module 2: Mathematical Models and Learning Algorithms

Indicative Content:

- Perceptron model
- Activation functions
- Gradient descent

Mapped Standards / Frameworks:

- ISO/IEC 42001 – Risk and lifecycle controls
- NIST AI RMF – Measure

Module 3: Multilayer Networks and Deep Learning

Indicative Content:

- Backpropagation
- Deep architectures
- Training challenges

Mapped Standards / Frameworks:

- NIST AI RMF – Manage
- ISO/IEC 42001 – AI system operation

Module 4: Specialized Architectures and Applications

Indicative Content:

- CNNs
- RNNs and LSTMs
- Industry use cases

Mapped Standards / Frameworks:

- NIST AI RMF – Map
- ISO/IEC 42001 – Operational controls

Module 5: Model Evaluation and Explainable AI

Indicative Content:

- Overfitting
- Model evaluation metrics
- Explainability techniques

Mapped Standards / Frameworks:

- ISO/IEC 42001 – Performance evaluation
- NIST AI RMF – Measure

Module 6: AI Ethics, Governance, and Responsible AI

Indicative Content:

- Bias and fairness
- Accountability
- AI governance frameworks

Mapped Standards / Frameworks:

- ISO/IEC 42001 – Responsible AI
- NIST AI RMF – Govern

List of Assessments and Marks

Student performance in this course shall be evaluated through a combination of continuous assessment and a final examination, structured as follows:

- **Continuous Assessment Tests (CATs) and Assignments: 30%**
  These shall assess students' understanding of theoretical concepts, practical application of skills, problem-solving ability, and alignment with prescribed standards and frameworks covered in the course.

- **Final Examination: 70%**
  The final examination shall evaluate comprehensive knowledge of the course content, including conceptual understanding, analytical ability, and application of principles in real-world and professional contexts.

Key References (APA)

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press.

Russell, S., & Norvig, P. (2021). *Artificial intelligence: A modern approach* (4th ed.). Pearson.

ISO/IEC. (2023). *ISO/IEC 42001: Artificial intelligence management systems*.

NIST. (2023). *AI Risk Management Framework (AI RMF 1.0)*.